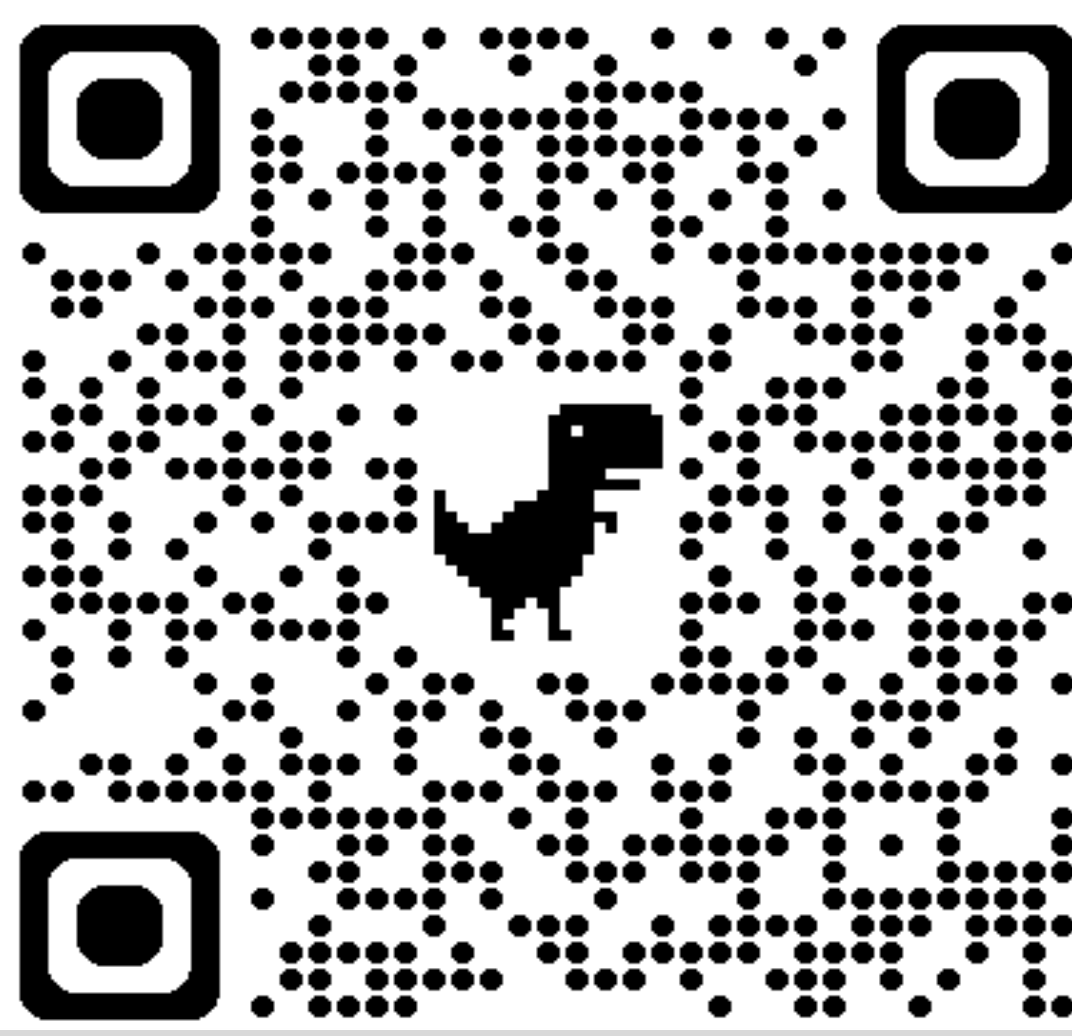


From Self-Play to Human Play: An Explainable Catan AI Platform

Play And Learn Catan Against The World’s Strongest Catan Bot

SCAN →
PLAY OUR BOT

<https://rlcatan.vercel.app/>
<https://github.com/AlgoCatan/RLCatan>



Settlers of Catan

A multiplayer strategy game where players gather resources, build, trade, and compete to reach 10 victory points.



Our Goal

Empower players of all skill levels to master Catan through *Experiential Learning* by providing a high-fidelity digital environment where they can practice, learn, and compete against sophisticated bots.

Our Motivation

Catan is the most popular modern board game, yet remains currently unsolved in the context of reinforcement learning. We hope to push the field forward with a competitive platform for Catan bots.

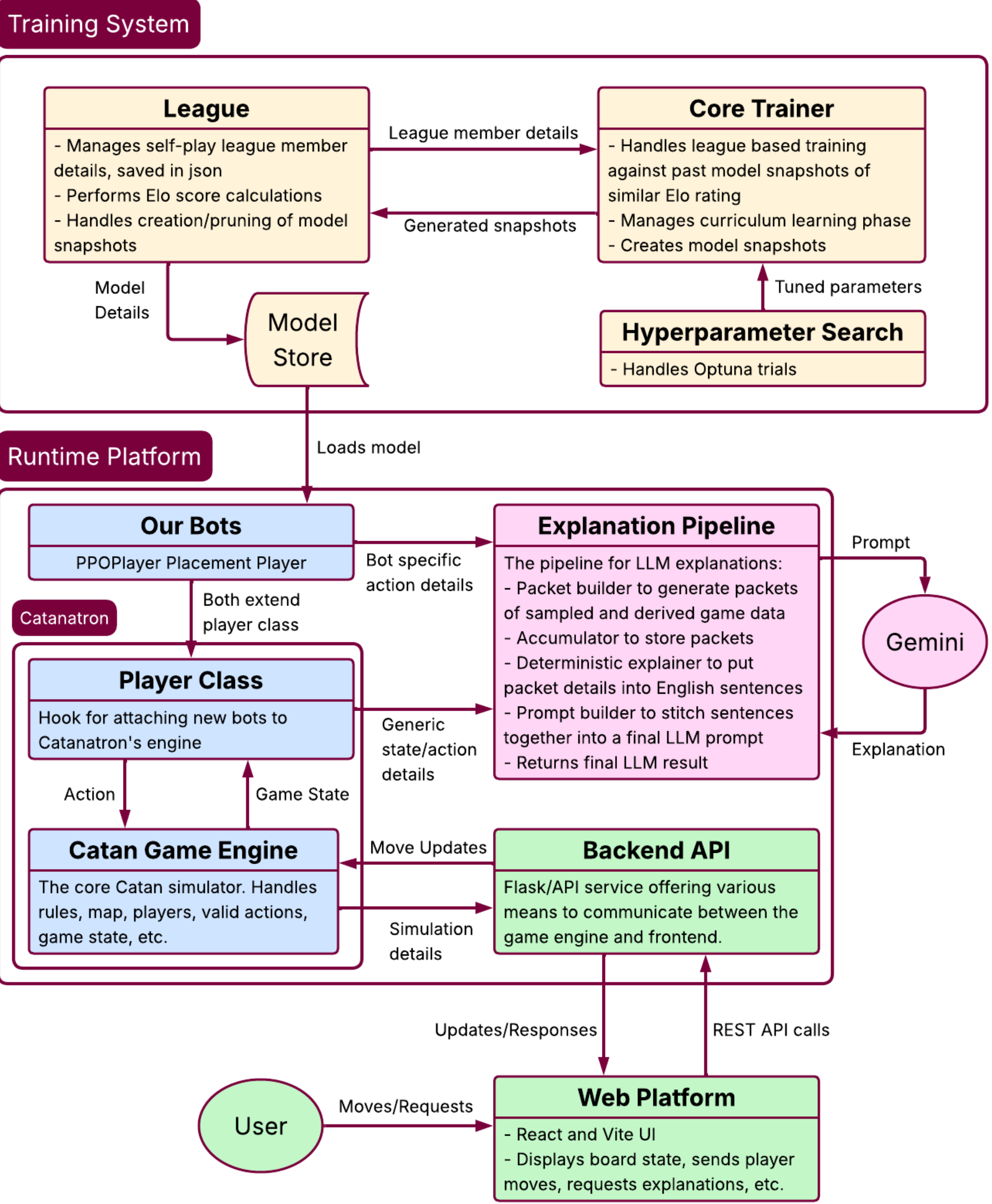
Plan to Achieve Goal

- Create an easy-to-use user interface.
- Train a bot to play Catan.
- Generate an explanation for every move.

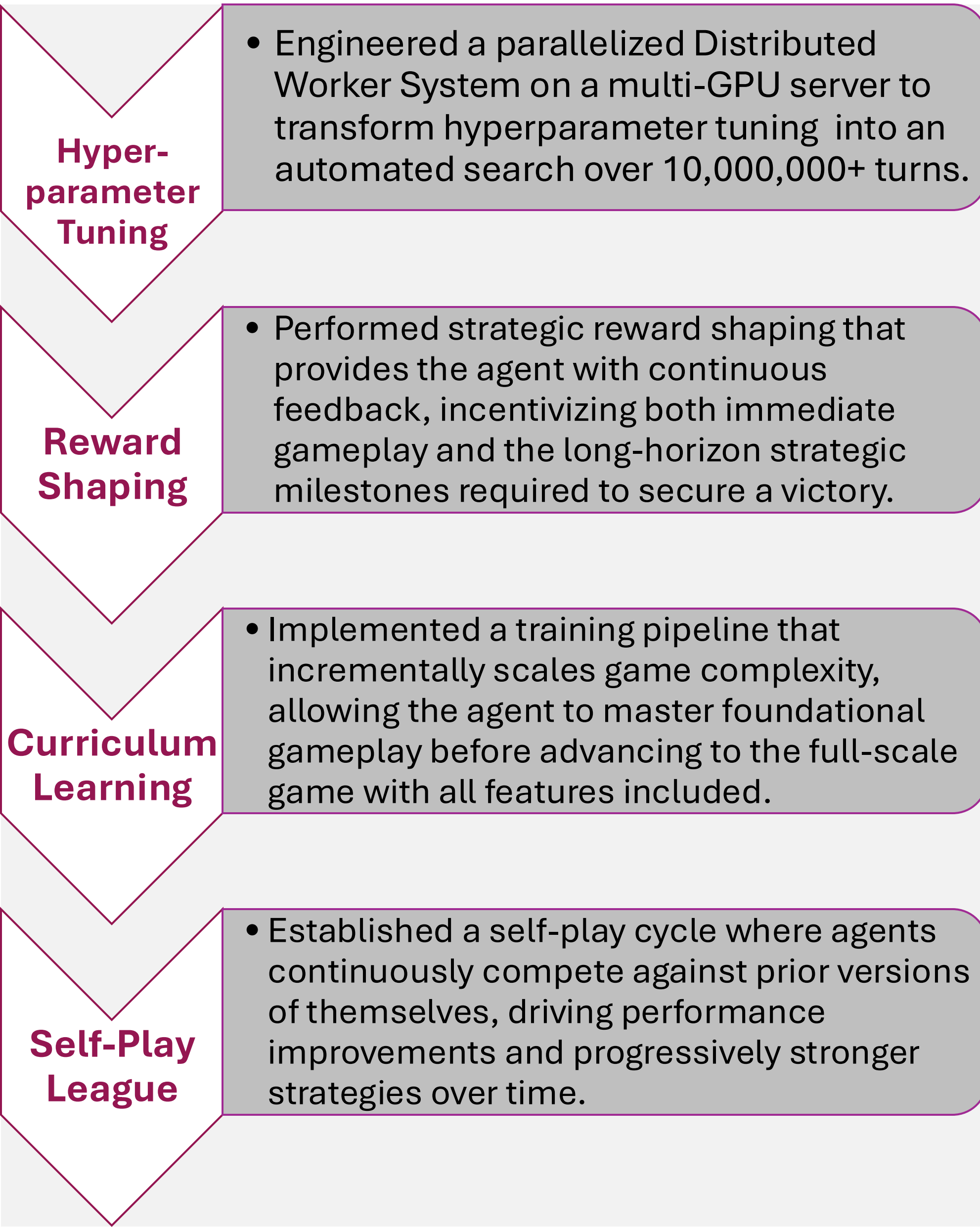
Interface



Architecture



PPO-Based Reinforcement Learning



Challenges/Accomplishments

Integration of Systems

- Designing a cohesive system that connects the Catanatron simulator, reinforcement learning pipeline, and LLM explanation layer required careful architectural decisions and custom environment wrappers to ensure seamless interaction between components.

Massive State & Action Space

- Catan is an extremely large state space with over 15 octillion possible boards and 290 actions. On top of that there is stochasticity because of the randomness involved. Efficiently exploring and learning within this space is a complex task. This is why our bot took weeks to train on 1,000,000+ games.

Usability

- Making the system interpretable and useful for humans introduced additional challenges. We developed LLM-generated explanations for each agent decision, alongside readability metrics (ex. Flesch scores) to ensure clarity and accessibility.

Team AlgoCatan



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5th year of Software Engineering



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5th year of Software Engineering



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Results

Matchup 1: Final Boss vs Baseline Random Bot

Final Boss (Our Bot)				Baseline Random Bot			
100 WINS				0 WINS			
Avg VP	Avg Settles	Avg Cities	Avg Ports	Avg VP	Avg Settles	Avg Cities	Avg Ports
15.09	4.73	4.00	0.18	2.68	1.87	0.18	0.18
Longest Road %	Longest Army %	Longest Road %	Longest Army %	Longest Road %	Longest Army %	Longest Road %	Longest Army %
99%	7%	1%	9%	1%	9%	1%	9%

Matchup 2: Final Boss vs Former World's Best Bot

Final Boss (Our Bot)				Former World's Best Bot			
60 WINS				40 WINS			
Avg VP	Avg Settles	Avg Cities	Avg Ports	Avg VP	Avg Settles	Avg Cities	Avg Ports
12.84	3.85	3.55	0.18	11.65	3.84	4.00	0.18
Longest Road %	Longest Army %	Longest Road %	Longest Army %	Longest Road %	Longest Army %	Longest Road %	Longest Army %
64%	11%	36%	15%	36%	15%	36%	15%

Tools

